Unit 6: Transformer and Machines

1. Transformers: Principle and Construction

Definition: A transformer is a static electrical device that transfers AC electrical power from one circuit to another at the same frequency, but usually at a changed voltage level (stepped up or stepped down).

Working Principle:

It works on the principle of **Faraday's Law of Mutual Electromagnetic Induction**. When an alternating voltage is applied to the primary winding, it creates a changing magnetic flux in the core. This alternating flux links with the secondary winding and induces an Alternating EMF in it.

Construction:

A basic transformer consists of two main parts:

- **Magnetic Core:** Made of laminated silicon steel to reduce eddy current and hysteresis losses. It provides a low reluctance path for the magnetic flux.
- **Windings:** Two sets of copper coils (Primary and Secondary) which are electrically isolated but magnetically linked.

Types based on Construction:

- **Core Type:** The windings surround a considerable part of the core. Used for high voltage, low output applications.
- **Shell Type:** The core surrounds a considerable portion of the windings. Provides better mechanical protection and is used for low voltage, high output applications.

👉 **Hinglish Explanation:** Transformer ek aisi machine hai jisme koi ghumne wala part (static) nahi hota. Ye AC voltage ko kam (Step-down) ya zyada (Step-up) karta hai, par **Frequency hamesha same rehti hai**. Ye Mutual Induction pe kaam karta hai—ek coil me current doge toh dusri me apne aap voltage ban jayegi. Iski core 'laminated' silicon steel se banti hai taaki losses kam hon.

2. EMF Equation and Transformation Ratio

Definition: The EMF equation relates the induced voltage in the transformer windings to the

frequency, number of turns, and maximum magnetic flux. The Transformation ratio (K) defines the relationship between the primary and secondary parameters.

EMF Equation:

Let N_1 = No. of turns in primary, f = frequency in Hz, and Φ_m = maximum flux in the core (in Webers).

Primary EMF: $E_1 = 4.44 \cdot f \cdot N_1 \cdot \Phi_m$
Secondary EMF: $E_2 = 4.44 \cdot f \cdot N_2 \cdot \Phi_m$

Transformation Ratio (K):

It is the ratio of secondary voltage to primary voltage, which is directly proportional to the turns ratio and inversely proportional to the current ratio.

$$K = \frac{V_2}{V_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$$

- If **K > 1** ($N_2 > N_1$): Step-Up Transformer (Voltage increases).
- If **K < 1** ($N_2 < N_1$): Step-Down Transformer (Voltage decreases).

Note: Since Power input $\approx$ Power output ($V_1 I_1 = V_2 I_2$), an increase in voltage results in a proportional decrease in current.

👉 **Hinglish Explanation:** EMF equation ka formula exam me direct derivation ya numerical me aata hai ($4.44 f N \Phi_m$). **Transformation ratio (K)** batata hai ki transformer voltage ko kitna guna badhayega ya ghatayega. Dhyan rahe, agar voltage badhti hai (Step-Up), toh current ussi hisaab se kam ho jata hai taaki total Power same rahe.

3. Auto Transformers

Definition: An auto-transformer is a special type of transformer that has only one single continuous winding which serves as both the primary and secondary.

Working Principle:

Unlike a standard two-winding transformer (which works on mutual induction), an auto-transformer works on the principle of **Self-Induction** as well as mutual induction. A portion of the single winding is shared between the primary and secondary circuits.

Advantages:

- **Less Copper Required:** Since a part of the winding is common, the total weight and cost of copper are reduced significantly.
- **Higher Efficiency:** Lower copper losses (I^2R) and smaller core size lead to higher overall efficiency.
- **Variable Output Voltage:** By using a sliding contact (like in a Variac), a continuously variable AC voltage can be obtained.

Disadvantages:

There is **no electrical isolation** between the primary and secondary circuits. If the common part of the winding breaks, full primary voltage may appear across the secondary, which is highly dangerous.

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AUTO-TRANSFORMER (Step-Down):

        Primary (V1)
       +-----------+
       |           |
      ===        +---+
      | |        |   |
      | |<-------+   | Secondary (V2)
      | |        |   |
      | |        +---+
      ===          |
       |           |
       +-----------+
```

👉 **Hinglish Explanation:** Normal transformer me 2 alag taar (windings) hote hain, par **Auto-transformer me sirf 1 hi lamba taar hota hai**. Usi me se thoda hissa primary aur thoda secondary ka kaam karta hai. Fayda ye hai ki copper ka kharcha bach jata hai aur efficiency achhi milti hai (jaise lab ka 'Variac' jisse voltage kam-zyada karte hain). Nuksan ye hai ki agar taar beech se tut gaya, toh current sidha aage chala jayega aur device jal sakta hai.

4. Motors: Construction and Working Principle

Definition: An electric motor is an electromechanical machine that converts electrical energy into mechanical energy.

Working Principle (DC Motor):

It works on the principle that **when a current-carrying conductor is placed in a magnetic field, it experiences a mechanical force**. The magnitude of this force is given by $F = B I L \sin(\theta)$, and its direction is determined by **Fleming's Left-Hand Rule**.

Construction Details:

- **Stator / Yoke:** The stationary outer frame that provides mechanical protection and carries the magnetic flux produced by the poles.
- **Poles & Field Windings:** Electromagnets fixed to the inner wall of the yoke to produce the main magnetic field.
- **Armature (Rotor):** The rotating part inside the motor. It consists of a cylindrical iron core with slots that carry the armature copper windings.
- **Commutator:** A split-ring copper device mounted on the shaft. It reverses the direction of current in the armature coils as they cross the magnetic poles, ensuring unidirectional continuous rotation.
- **Brushes:** Made of carbon, these rest against the commutator to supply external electrical current to the rotating armature.

👉 **Hinglish Explanation:** Motor bijli khati hai aur mechanical power (ghumna) deti hai. Ye is rule pe chalti hai ki jab kisi current wale taar ko magnet ke beech rakhte hain toh uspar dhakka (force) lagta hai. Direction nikalne ke liye **Fleming ka Left-Hand Rule** use karte hain. **Commutator** motor ka sabse special part hai; ghumte hue armature ko current dene aur uski direction theek rakhne ka kaam yahi karta hai.

5. Basic Equations and Characteristics of Motors

Definition: The behavior of a motor is governed by mathematical equations relating voltage, current, and speed. Motor characteristics graphically show how speed and torque change with respect to armature current.

1. Back EMF (E_b):

When the motor armature rotates in the magnetic field, an EMF is induced in it according to Faraday's law. By Lenz's law, this EMF opposes the applied supply voltage (V). It is called Back EMF or Counter EMF. It regulates the armature current.

$$E_b = \frac{P \Phi Z N}{60 A}$$

Where P=Poles, Φ=Flux per pole, Z=Total conductors, N=Speed in RPM, A=Parallel paths.

2. Voltage Equation of a DC Motor:

The applied voltage V must overcome the Back EMF and the voltage drop in the armature resistance (R_a).

$$V = E_b + I_a R_a$$

Where I_a is the Armature Current.

3. Characteristics of DC Motors:

- **DC Series Motor:** The field winding is in series with the armature.
  Characteristics: Very high starting torque. Speed drops drastically as load increases. Never started without a load.
  Applications: Electric trains, cranes, hoists.
- **DC Shunt Motor:** The field winding is in parallel with the armature.
  Characteristics: Speed remains almost constant regardless of load variations. Moderate starting torque.
  Applications: Lathe machines, fans, centrifugal pumps.

👉 **Hinglish Explanation:** Jab motor ghumti hai toh uske andar ek voltage banti hai jo main supply ko oppose karti hai, ise **Back EMF (E_b)** kehte hain. Ye bahut zaroori hai warna motor jal jayegi. Voltage equation $V = E_b + I_aR_a$ hoti hai. **Series motor** shuruat me bahut taqat lagati hai (High Starting Torque), isliye train me use hoti hai. Par **Shunt motor** ki speed fix rehti hai (Constant speed), isliye fans aur lathe machines me use hoti hai.

🔁 Quick Revision Notes (Exam Saver)

- **Transformer:** Static device. Changes voltage (step-up/down). Works on Mutual Induction. **Frequency is constant.**
- **Transformation Ratio (K):** $K = V_2/V_1 = N_2/N_1 = I_1/I_2$.
- **Auto Transformer:** Only 1 winding. Saves copper but provides no electrical isolation. Works on self and mutual induction.

- **Motor Principle:** Electrical to Mechanical energy. Current carrying conductor in magnetic field gets a force ($F=BIL$). Fleming's Left-Hand rule.
- **Back EMF (E_b):** Opposes supply voltage (V). Keeps motor safe. Equation: $V = E_b + I_a R_a$.
- **Motor Types:** Series motor = High starting torque (Trains). Shunt motor = Constant speed (Fans/Lathes).

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